

Phase B — Fake-league characterization (not curve fitting)

Knobs (mini glossary)

Symbol	Plain name	What it controls
$\hat{A}_i$	Player ability	PPM within season
$\hat{T}$	Realized team talent	Mean $\hat{A}_i$ on roster, not T_{j*}
T_{j*}	Sim assignment target	Pass B only – not observed in NCAA data
ρ	Assignment assortativity	Sim soft-match knob; overlay forensics on data
L_C	Team congestion	$mean_j \sigma(\gamma(\hat{A}_i - \theta))$ per roster
λ	Congestion weight	Score $S_i = A_i - \lambda \cdot L_C$ (sim / later on data)
θ	Viability cutline	$F_{\hat{A}}^{-1}(1 - K/N)$ on empirical $\hat{A}_i$
γ	Sigmoid sharpness	Reshapes team L_C – 539 placeholder until lock
K/N	Draft rate	Accepted / total in filtered panel ($\approx .1.8\%$)

Deck map (assign →score)

- Slide 2 — $\hat{A}_i, \hat{T}_j$ inputs
- Slide 3 — team interval overlap (ρ diagnostic, 530 CELL 8)
- Slide 4 — team L_C distribution
- Slide 5 — Sketch A: $\hat{T}_j$ vs L_C
- Slide 6 — γ sweep on real rosters
- Slide 7 — sim ρ vs empirical coverage (optional add)

Claim (PD17): Map empirical properties of real rosters — prerequisite for pinning ρ and θ from data, then γ and λ without hero-curve matching.

PD17 — Empirical MBB inputs: $\hat{A}_i$ and $\hat{T}_j$

MBB 2011-2021 · ppm z within season · min 20 min · poolq winsor 0.01–0.99

Realized rosters — descriptive inputs, not sim draws.

Left: $\hat{A}_i$ — player ability (one value per player-season).

Right: $\hat{T}_j$ — mean $\hat{A}_i$ on roster (team-season; not LOO).

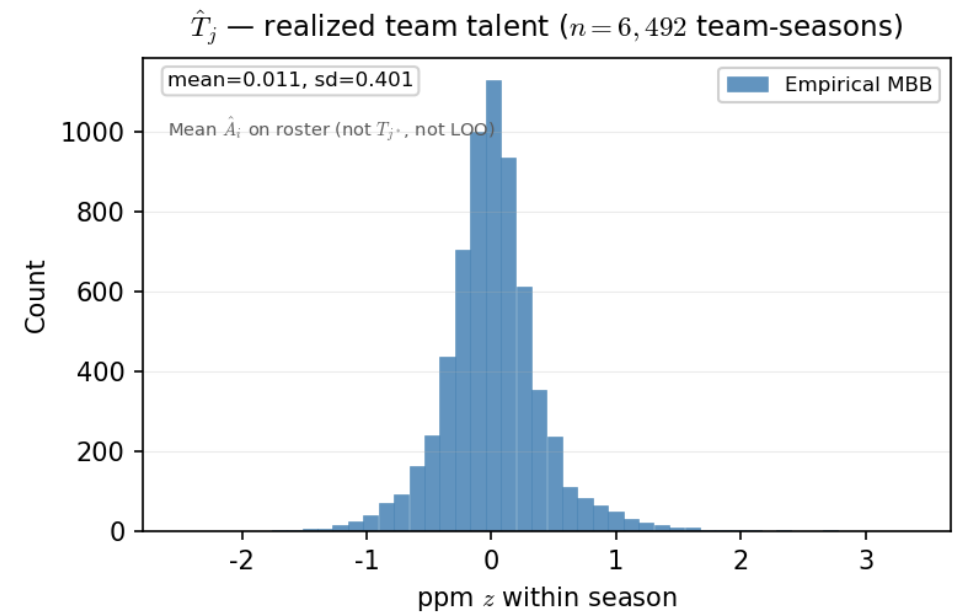
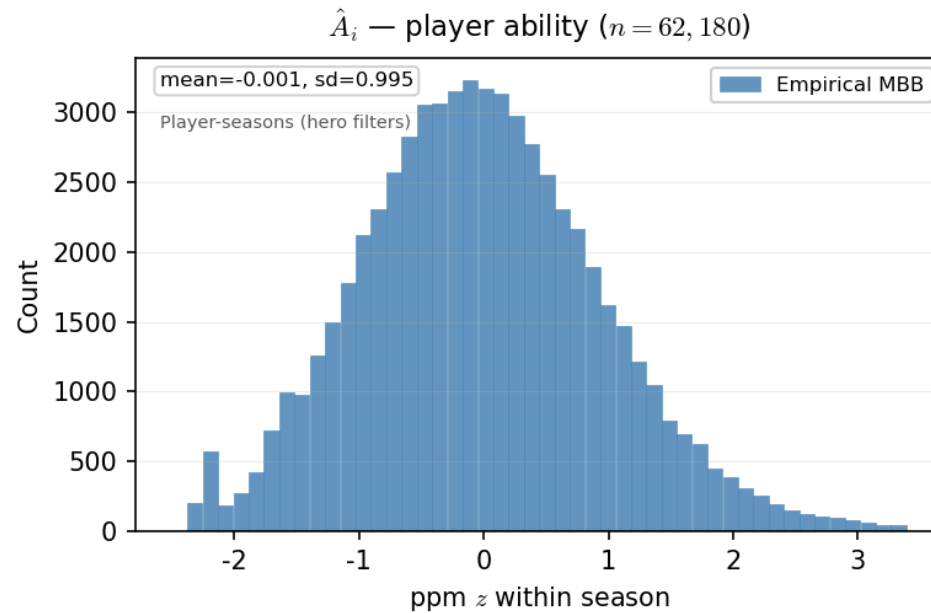
$\hat{T}_j$ is tighter than $\hat{A}_i$ (averaging shrinks spread).

This run: mean $\hat{A}_i$ =-0.001, mean $\hat{T}_j$ =0.011 (sd 0.995 / 0.401).

PD17 θ proxy: $K/N = 0.0182$.

(1,134 drafted / 62,180 player-seasons).

Empirical MBB inputs — $\hat{A}_i$ and $\hat{T}_j$ (2011-2021, min 20 min, poolq winsor 0.01–0.99)



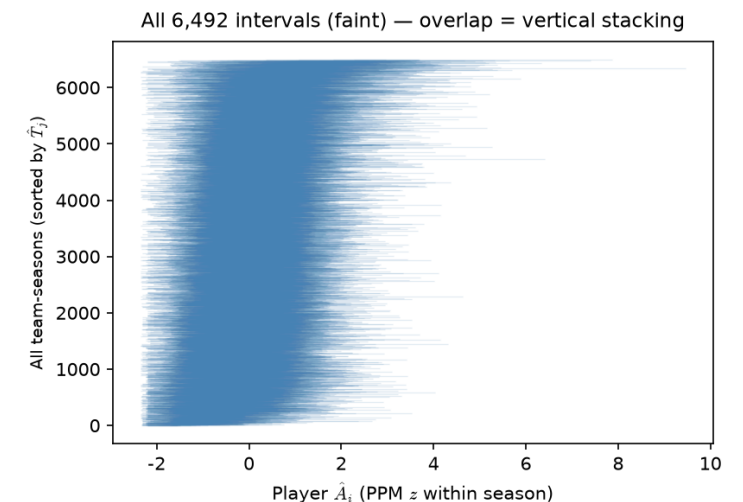
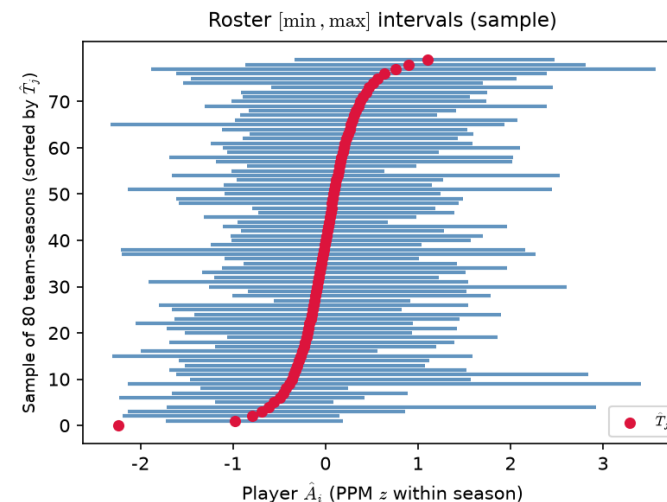
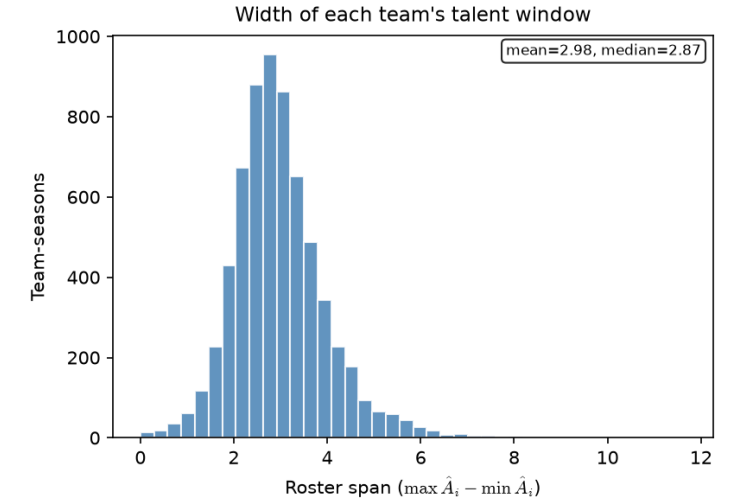
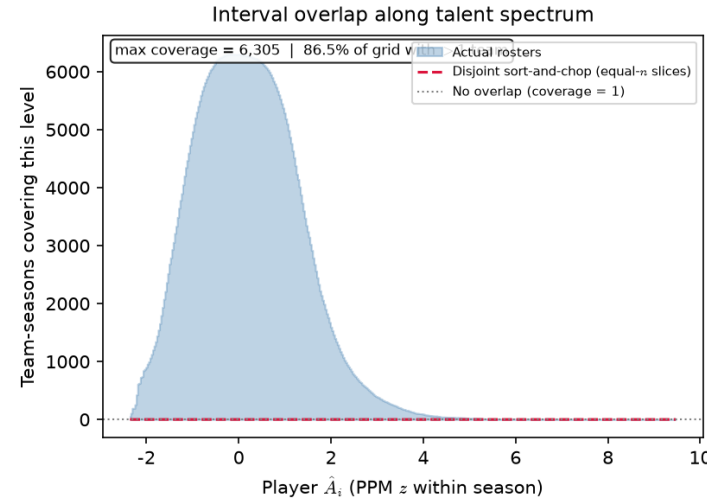
Claim (PD17): World-first inputs from the real MBB panel — $\hat{A}_i$ per player-season and $\hat{T}_j$ per team-season. Not T_{j*} (sim assignment targets).

PD17 — Empirical team $\hat{A}_i$ interval overlap (ρ diagnostic)

MBB 2011-2021 · PPM z within season · min 20 min · poolq winsor 0.01–0.99 · 6,492 team-seasons (530 CELL 8 port)

- Each team-season: talent window $[\min \hat{A}_i, \max \hat{A}_i]$ on PPM z.
- Coverage = how many windows cover a point on the spectrum (530 CELL 8).
- Red dashed: disjoint sort-and-chop on same player-seasons (537 B analog).
- Actual max coverage=6,305; sort-and-chop max=7.
- 86% of grid points have > 1 team covering.
- Roster span: mean=2.98 z, median=2.87 z (6,492 team-seasons).
- Pairs with Phase B ρ slide — sim ρ dials overlap between these curves.

Do team talent windows overlap on real rosters?
530 CELL 8 forensics — empirical ρ diagnostic (PD17)



Claim (PD17): Real NCAA rosters leave massively overlapping team talent windows — the empirical target ρ was meant to match in sim (530 CELL 8 / 538 Plot A).

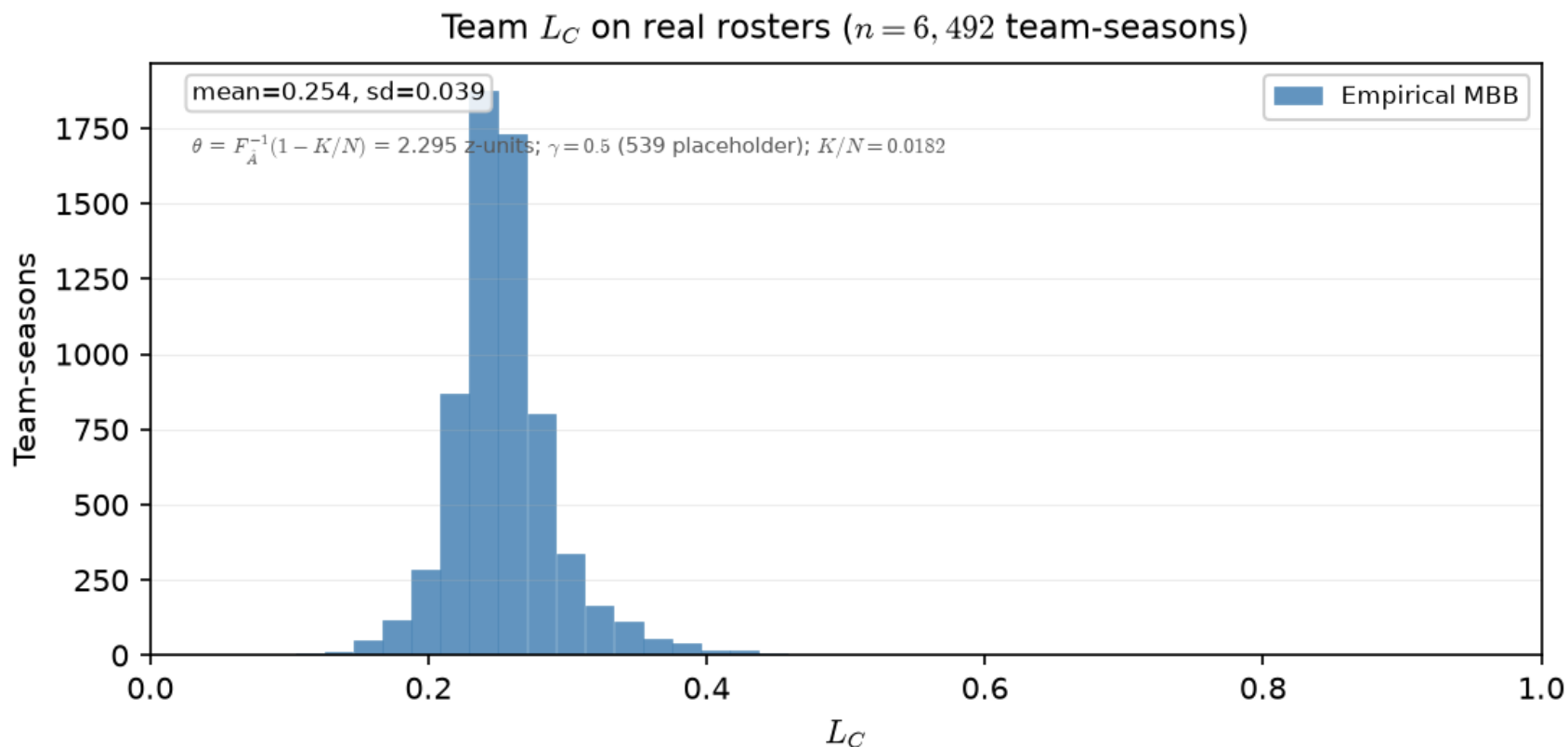
PD17 — Empirical team L_C distribution

Given overlapping talent windows (prior slide), team L_C summarizes peer viability on each roster.

MBB 2011-2021 · team smooth L_C · PPM z within season · min 20 min · $\theta = F_A^{-1}(1 - K/N) = 2.295$ ($K/N = 0.0182$) · $\gamma = 0.5$ placeholder

- Team $L_C = \text{mean}_j \sigma(\gamma(\hat{A}_j - \theta))$ on each roster.
- Same PD16 team-smooth definition — one value per team-season.
- Not LOO; includes every roster member in the mean.
- $\gamma = 0.5$ (539 placeholder until PD17 γ / λ calibration).
- This run: mean $L_C = 0.254$, sd = 0.039 (6,492 team-seasons).
- θ from draft rate: $K/N = 0.0182 \Rightarrow \theta = 2.295$ on $\hat{A}_i$.

Empirical MBB — team smooth L_C (2011-2021, min 20 min, poolq winsor 0.01-0.99)



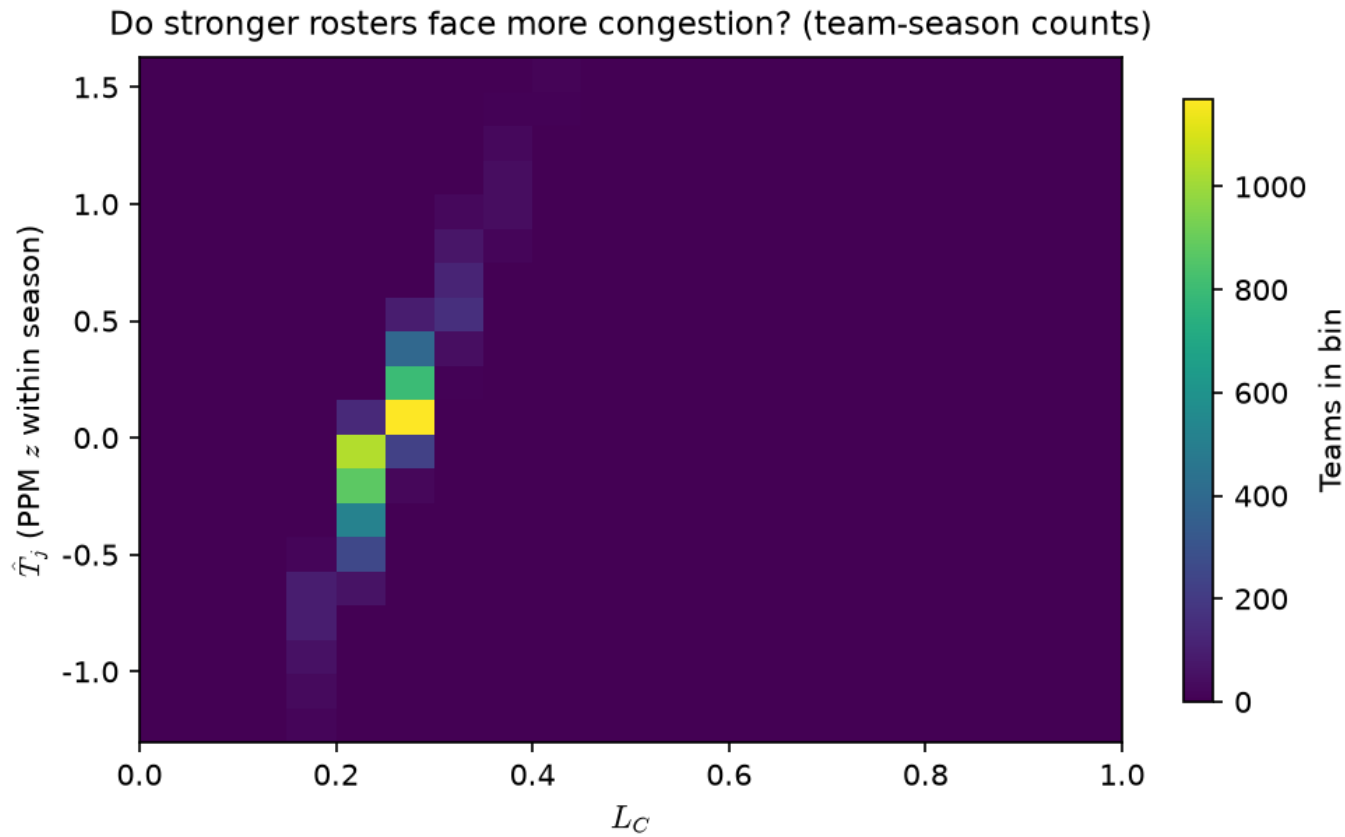
Claim (PD17): Real rosters carry a spread of team congestion L_C — the descriptive input before we pin ρ and γ in sim.

PD17 — Empirical Sketch A: $\hat{T}_j$ vs L_C

MBB 2011-2021 · team smooth L_C · PPM z within season · min 20 min · $\theta = F_A^{-1}(1 - K/N) = 2.295.(K/N = 0.0182)$ · $\gamma = 0.5$ placeholder

- Each cell: count of team-seasons at $(\hat{T}_j, L_C)$.
- Color = count (empirical Sketch A — no ρ arms).
- $\hat{T}_j$ = mean $\hat{A}_i$ on roster (realized talent, not T_{j*}).
- Look for upward co-movement: higher $\hat{T}_j$ with higher L_C .
- Panel: mean $\hat{T}_j = 0.011$, mean $L_C = 0.254$.

Empirical Sketch A — $\hat{T}_j$ vs L_C ($\theta = 2.295$ z-units, $\gamma = 0.5$)

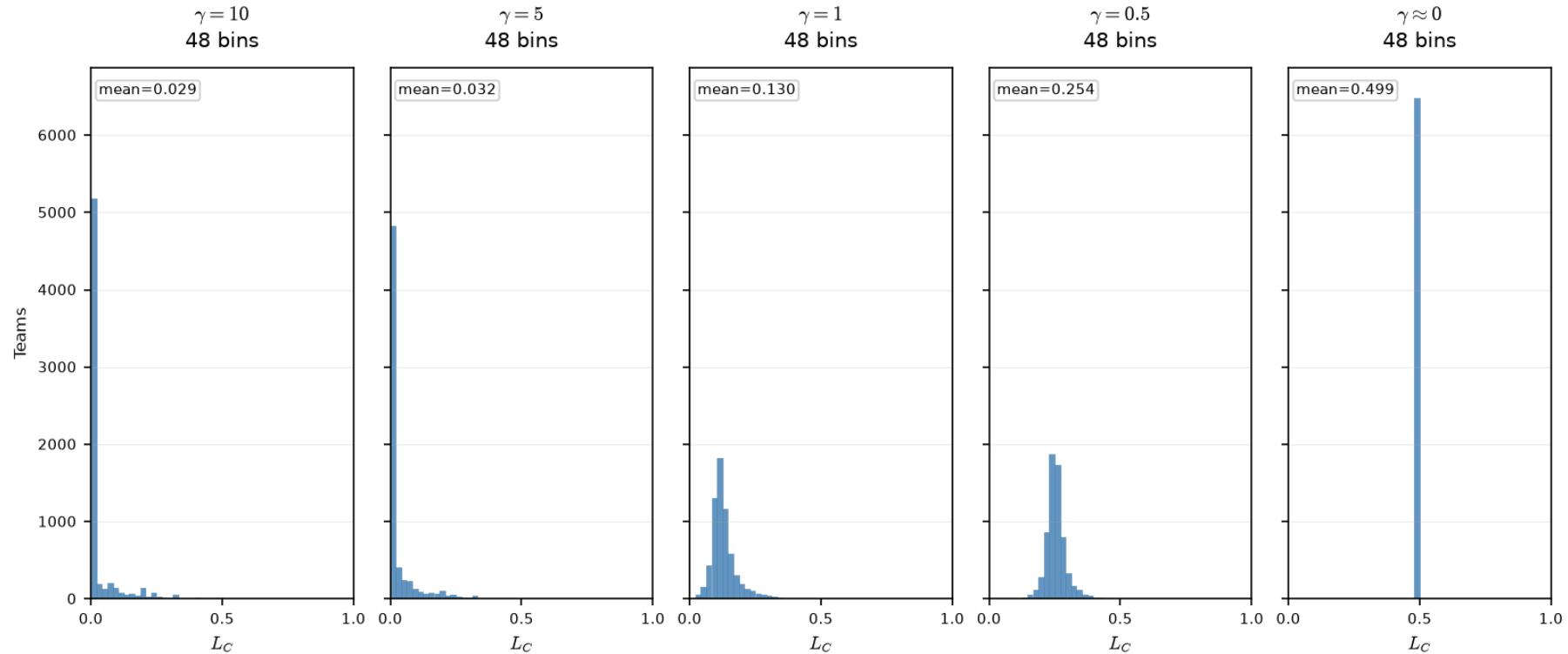


Claim (PD17): On actual team-seasons, stronger realized rosters ($\hat{T}_j$) tend to co-occur with higher L_C — empirical Sketch A.

PD17 — Empirical γ sweep: team L_C on real rosters

MBB 2011-2021 · team smooth L_C · PPM z · min 20 min · $\theta = 2.295$ z-units ($K/N = 0.0182$) fixed across γ arms

Empirical MBB — team L_C vs γ ($\theta = 2.295$ z-units, $K/N = 0.0182$)



OAT on real panel: fixed θ (K/N cutline on PPM z); only γ in $\sigma(\gamma(\hat{A} - \theta))$ changes.

Arms: $\gamma \in \{10, 5, 1, 0.5, 0.001\}$ — same team-seasons, same rosters.

At $\gamma = 10$: mean $L_C = 0.029$, 84% teams below 0.05.

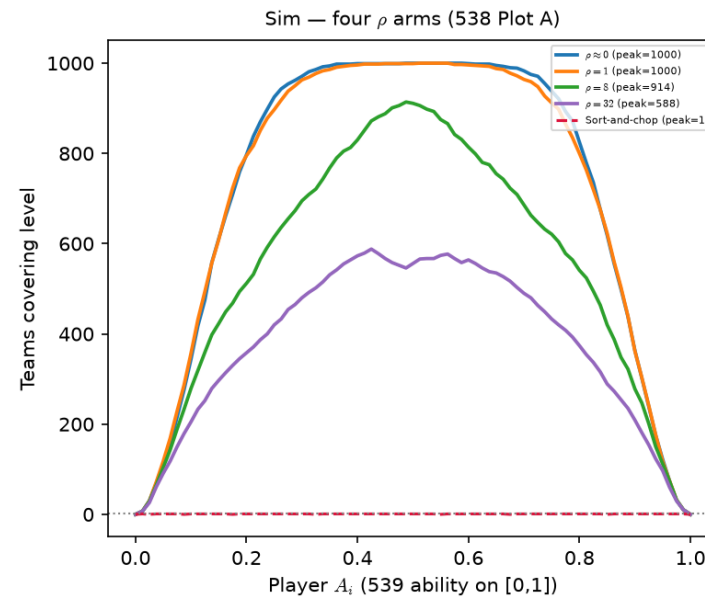
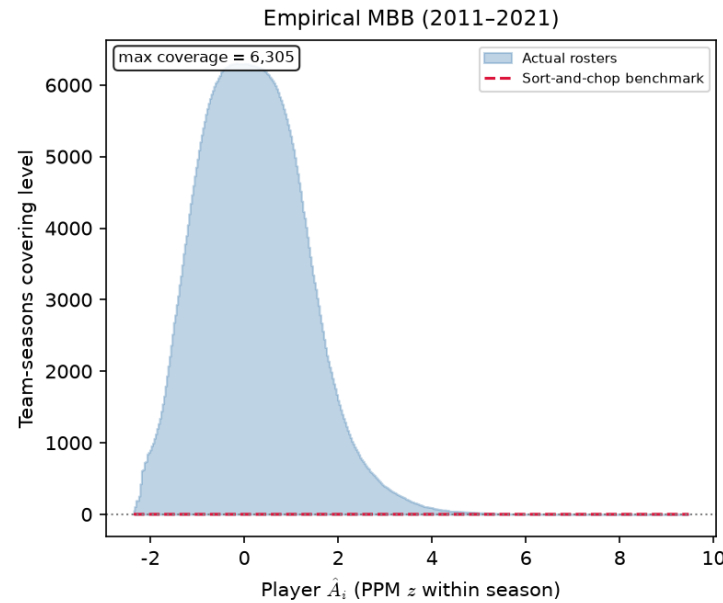
At $\gamma \approx 0$: mean $L_C = 0.499$, max $L_C = 0.500$ — spread opens on z-scale.

Claim (PD17): On real rosters, γ reshapes team L_C — high γ compresses viability near 0 on the PPM-z scale; low γ spreads L_C into the interior of $[0,1]$.

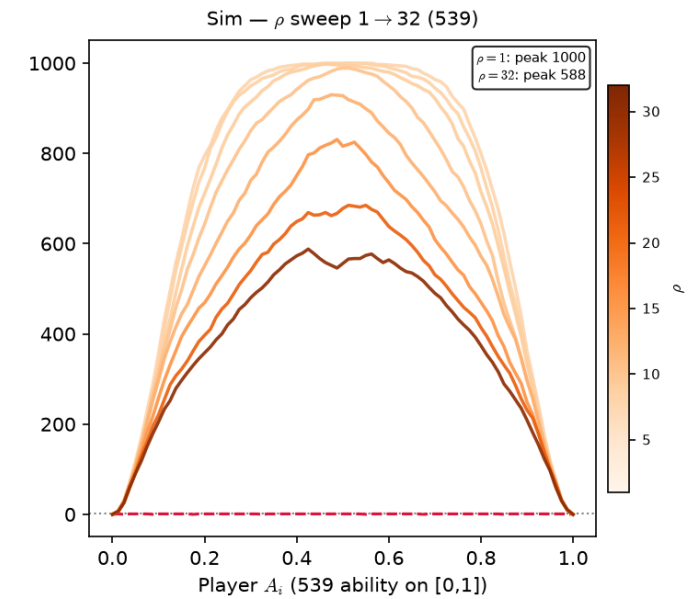
PD17 — Empirical vs sim interval overlap (\rho calibration target)

Assign block capstone · empirical MBB vs 539 sim · same coverage diagnostic

Interval overlap: empirical target vs sim ρ (four arms + 1 $\rightarrow$ 32 sweep)



Axes not overlaid — PPM z (empirical) vs [0,1] ability (sim). Compare coverage peak and shape.



- Three panels: empirical (left) · four ρ arms (center) · $\rho = 1 \rightarrow 32$ sweep (right).
- Same A_i / T_j^* draw across sim panels; seed fixed.
- Empirical max coverage = 6,305; sort-and-chop max = 7.
- Center: $\rho \in \{0.001, 1, 8, 32\}$; right sweep peaks $\rho = 1 \rightarrow 1000$, $\rho = 32 \rightarrow 588$.
- Sim sort-and-chop peak = 1 on both sim panels.
- PPM z vs [0,1] ability — compare shape, seed 42.

Claim (PD17): Real rosters sit at high coverage peak — sim ρ must be tuned so soft assign matches that overlap, not sort-and-chop.